

Partial EHL analysis of rib–roller end contact in tapered roller bearings

W. Wang*, P. L. Wong[†] and Z. Zhang*

A partial EHL analysis was performed for the tapered rib/spherical roller end contact in tapered roller bearings. The average Reynolds equation, the elasticity equation and the pressure–viscosity relation were solved simultaneously. The effects of the surface roughness as well as the peculiar geometrical and kinematics parameters of the rib–roller end contact on the friction torque and film thickness were investigated. The optimal ratios of radius of curvature of roller end to rib face were deduced, which confirm the previous finding with the theory of smooth surfaces. The significant range of surface roughness, and the optimal surface roughness for the roller big end were obtained. It was found that asperity contacts extend into the outlet zone. The results are significant for the design of rib faces and roller ends. The theoretical treatment is validated by its good correlation with the existing experimental data for smooth surface contact. Copyright © 1996 Elsevier Science Ltd

Keywords: tapered roller bearing, partial elastohydrodynamic lubrication, surface roughness

Introduction

Tapered roller bearings are widely used in automobiles and industrial machinery because they have the ability to carry not only heavy radial loads but also axial and combined loads. Due to the conical shape of the rollers, the forces acting on a roller from the inner and outer races differ slightly in direction, their resultant pushes the roller towards its bigger end and friction is thus induced between the rib surface and roller end. For heavy load or high speed conditions, lubrication problems in rib/roller end contact can be serious. Hence, many investigations of friction, film

thickness and lubrication of rib/roller end contacts have been conducted^{1–6}.

Korrenn¹ started the study of the friction of rib/roller end contacts by carrying out experiments. The effects of geometric configuration and position of nominal contact point to favour lubrication were found by Jamison *et al.*². Dalmaz *et al.*^{3,4} determined the optimal geometric configuration of rib/roller end contact by using an optical simulator rig and simulated the results with a hydrodynamic approach. It was pointed out that the elastic deformation of surfaces has to be considered in the theoretical study. With the development of elastohydrodynamic lubrication (EHL) theory, Zhang *et al.*⁵ started an EHL analysis of the rib/roller end contact with the inclusion of special geometry and kinematics conditions, assuming Newtonian lubricant and isothermal conditions. Recently, non-Newtonian and thermal effects were included in an EHL study by Jiang *et al.*⁶. It was found that under the investigated working conditions the non-Newtonian effect is very minimal and the thermal effect is somehow greater but only noticeable at high speed.

Correspondence should be addressed to Mr Wang Wen, PO Box 224, Department of Mechanical Engineering, Shanghai University, 149, Yan-Chang Road, Shanghai, P.R. China

* Department of Mechanical Engineering, Shanghai University, 149 Yan-Chang Road, Shanghai 200072, P.R. China

[†] Department of Manufacturing Engineering, City University of Hong Kong, 83 Tat Chee Avenue, Kowloon, Hong Kong

Received 20 March 1995; revised 25 September 1995; accepted 24 November 1995

Notation			
a, b	semiminor and semimajor axes of contact ellipse respectively (m)	n	rotation speed (nps)
K	ellipticity parameter, $K = a/b$	u_s	$0.5(u_{1(0)} + u_{2(0)})$, average entraining velocity (m/s)
h_T	nominal film thickness (m)	U_1, U_2	dimensionless speed, u_1/u_s and u_2/u_s
hc	nominal central film thickness (m)	W_1/W_2	dimensionless speed, w_1/u_s w_2/u_s
hm	nominal minimum film thickness (m)	E_1, E_2	Young's moduli (N/m ²)
hsm	nominal minimum film thickness for smooth surface contact (m)	μ_1, μ_2	Poisson's ratios
h_0	$hc - \delta$ (0,0) (m)	E	effective elastic modulus
δ	elastic deformation (m)	p	pressure (N/m ²)
Δ_1, Δ_2	the roughness of contact surface (m)	$\bar{p}$	mean hydrodynamic pressure (N/m ²)
σ_1, σ_2	surface roughness standard deviation (m)	$\bar{P}$	nondimensional mean pressure, p/E
σ	composite roughness standard deviation (m)	$\bar{P}_a$	nondimensional asperity contact pressure
r_i, r_0	inner and outer radii of rib (m)	λ	h_T/σ
r	large end radius of roller (m)	$\bar{\sigma}$	σ/Rx
Rx, Rz	equivalent radii of curvature in x and z (m)	N	number of asperities per unit area
R_{x1}, R_{x2}	radii of curvature in x of rib and of roller end respectively (m)	β	mean radius of curvature of asperity
β_1	half cone angle	W_T	total dimensionless load, $W_T/(E R_x^2)$
Ψ	angle formed by bearing axis and normal line of rib at nominal contact point	W_h	hydrodynamic load
ϵ	distance from the nominal contact point to inner race (m)	W_a	asperity contact load
X	nondimensional coordinate in x , $X = x/b$	τ	shear stress
Z	nondimensional coordinate in z , $Z = z/a$	$\bar{\eta}$	dimensionless viscosity, η/η_0
Y	nondimensional coordinate in film thickness direction, $Y = y/b$	η_0	ambient viscosity of lubricant
H_T	nondimensional nominal film thickness h_T/Rx	τ_c	equivalent shear stress
		$F(\tau_c)$	rheological function, for Newtonian, = 1
		M	friction torque acting on rib around bearing axis (N m)
		$M_{r,a}$	asperity contact friction torque
		$M_{r,h}$	hydrodynamic friction torque
		α	half roller angle
		h_T	average gap
		q_x, q_z	flow in x and z direction
		$\bar{q}_x, \bar{q}_z$	average flow

However, all previous research, both experimental and theoretical, was conducted with the assumption of perfectly smooth surfaces. In practice, the tapered roller bearings often operate with minimum film thickness of the order of a few tenths of a micron. Furthermore, the surface roughnesses of roller ends and rib surface are poorer than that of rollers and raceway surfaces. It has been pointed out by Kleckner and Pirvics⁷ that the dimensions of the asperities of roller ends and the rib surface are of the same order of magnitude as the built up oil film. Figure 1 shows a typical surface roughness measurement of the roller end of a tapered roller bearing. It can be seen that the magnitude of roughness is of the same order of film thickness as those which were obtained experimentally⁴ and theoretically⁶. In order to simulate as closely as possible the true physical situation, the surface roughness was thus included in the present analysis. Only one particular type of rib/roller end contact, tapered rib to spherical roller end, was selected in the study of roughness effects. This was not only because it is the most popular configuration but also it has the smallest film thickness. Hence, roughness effects are conspicuous with this configuration.

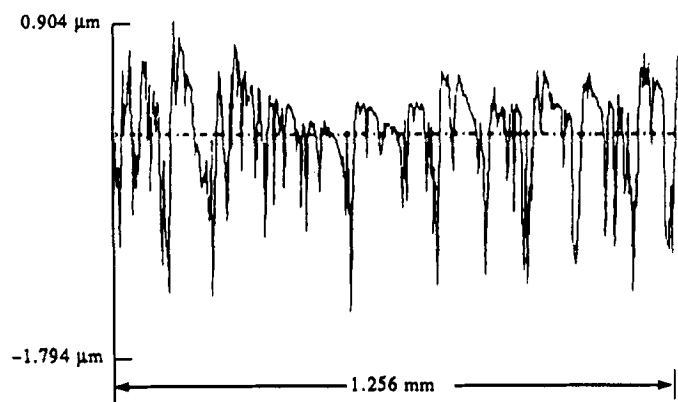


Fig 1 Roller end surface roughness $\sigma = 0.418 \mu\text{m}$

During the last two decades, a number of partial EHL analyses have been carried out taking into account the surface roughness effect on the film thickness. One of the major advances is to use a stochastic method to describe lubrication between rough surfaces. Utilizing the stochastic approach, Patir and Cheng⁸ put forward an average flow model for determining the effects of three-dimensional roughness on partial hydrodynamic

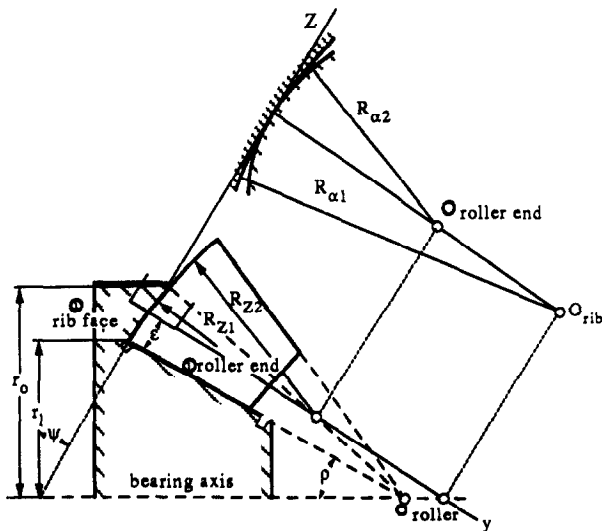


Fig 2 Geometry of rib-roller end contact

lubrication. The effects of surface roughness are expressed by some flow factors which can be obtained through numerical flow simulation and the model was confirmed by References 9–11. Hence, the average flow model was adopted in this work.

Theory

Geometry and kinematics

Figure 2 shows the rib/roller end contact geometry. It can be described by an equivalent sphere on a plane geometry as shown in Fig 3. For a tapered rib to spherical end contact, the nominal film shape can be given by:

$$h_T = h_0 + \frac{x^2}{2R_x^2} + \frac{z^2}{2R_z R_z} + \delta(x, z) \quad (1)$$

where $\delta(x, z)$ is the elastic deformation.

When under operation, rolling, sliding and spinning exist simultaneously, as illustrated in Fig 3. The velocities of the surfaces in the two principal directions are:

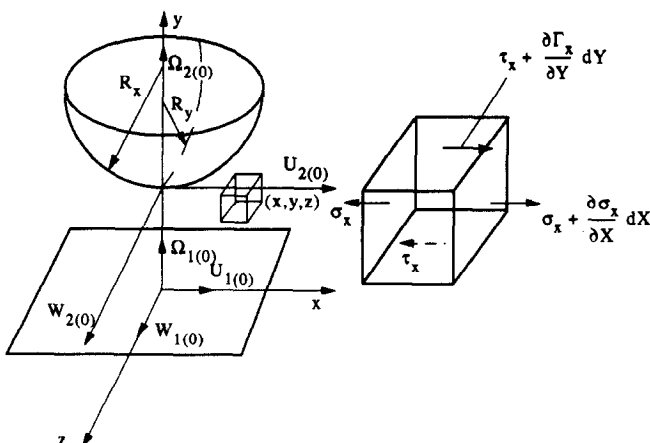


Fig 3 Analytical model

$$\begin{aligned} u_1(z) &= u_{1(0)} + z \cdot \Omega_{1(0)} \\ u_2(z) &= u_{2(0)} + z \cdot \Omega_{2(0)} \\ w_1(x) &= w_{1(0)} - x \cdot \Omega_{1(0)} \\ w_2(x) &= w_{2(0)} - x \cdot \Omega_{2(0)} \end{aligned} \quad (2)$$

Governing equations

Considering the force balance of a fluid element in the x -direction (Fig 3), it can be shown that:

$$\frac{\partial \tau_x}{\partial y} = \frac{\partial p}{\partial x}, \quad \frac{\partial u}{\partial y} = \frac{\tau_x}{\eta} F(\tau_e) \quad (3)$$

By integration:

$$\begin{aligned} u &= \frac{1}{\eta} \frac{\partial p}{\partial x} \int_0^y F(\tau_e) y dy \\ &+ \frac{u_2 - u_1 - \frac{1}{\eta} \frac{\partial p}{\partial x} \int_0^h F(\tau_e) y dy}{\int_0^h F(\tau_e) dy} \cdot \int_0^y F(\tau_e) dy + u_1 \end{aligned} \quad (4)$$

Hence, the flow rate per unit width of cross-section can be found as:

$$\begin{aligned} q_x &= \int_0^h u dy = \frac{1}{\eta} \frac{\partial p}{\partial x} \left[\int_0^h \int_0^y F(\tau_e) y dy dy \right. \\ &- \left. \frac{\int_0^h F(\tau_e) y dy}{\int_0^h F(\tau_e) dy} \int_0^h \int_0^y F(\tau_e) dy dy \right] \\ &+ \frac{u_2 - u_1}{\int_0^h F(\tau_e) dy} \int_0^h \int_0^y F(\tau_e) dy dy + \int_0^h u_1 dy \end{aligned} \quad (5)$$

With a control volume (dx, dz, h) selected as shown in Fig 4, the average flow in the x direction is expressed as:

$$\bar{q}_x = \frac{1}{\Delta Z} \int_z^{z+\Delta z} q_x dz \quad (6)$$

The absolute film thickness is:

$$h = h_T + \Delta_1 + \Delta_2 \quad (7)$$

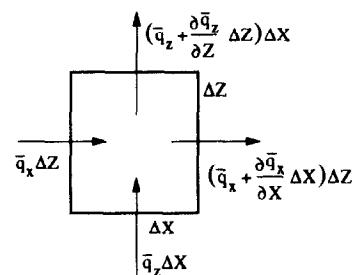


Fig 4 Control volume

and the average gap can be calculated as:

$$\bar{h}_T = \int_{-h_T}^{\infty} (h_T + \Delta) f(\Delta) d\Delta \quad (8)$$

where $\Delta = \Delta_1 + \Delta_2$. For a Gaussian surface¹¹, the function f is:

$$f(\Delta) = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left(-\frac{\Delta^2}{2\sigma^2}\right) \quad (9)$$

Equation (5) can be written in terms of pressure flow factors ϕ_x, ϕ_y and shear flow factors ϕ_{sx}, ϕ_{sz} , their definition being given by Patir and Cheng⁸. The mean flow can then be expressed as:

$$\begin{aligned} \bar{q}_x = & \phi_x \frac{1}{\eta} \frac{\partial \bar{p}}{\partial x} \left[\int_0^{h_T} \int_0^y F(\tau_e) dy dy \right. \\ & - \frac{\int_0^{h_T} F(\tau_e) dy}{\int_0^{h_T} F(\tau_e) dy} \cdot \int_0^{h_T} \int_0^y F(\tau_e) dy dy + \\ & \frac{u_2 - u_1}{\int_0^{h_T} F(\tau_e) dy} \int_0^{h_T} \int_0^y F(\tau_e) dy dy \\ & \left. + \int_0^{h_T} u_1 dy + \frac{u_2 - u_1}{2} \sigma \phi_{sx} \right] \quad (10) \end{aligned}$$

Similarly:

$$\begin{aligned} \bar{q}_z = & \phi_z \frac{1}{\eta} \frac{\partial \bar{p}}{\partial z} \left[\int_0^{h_T} \int_0^y F(\tau_e) dy dy \right. \\ & - \frac{\int_0^{h_T} F(\tau_e) dy}{\int_0^{h_T} F(\tau_e) dy} \int_0^{h_T} \int_0^y F(\tau_e) dy dy + \\ & \frac{w_2 - w_1}{\int_0^{h_T} F(\tau_e) dy} \int_0^{h_T} \int_0^y F(\tau_e) dy dy \\ & \left. + \int_0^{h_T} w_1 dy + \frac{w_2 - w_1}{2} \sigma \phi_{sz} \right] \quad (11) \end{aligned}$$

Considering the mean flow balance, we have:

$$\frac{\partial(\rho \bar{q}_x)}{\partial x} + \frac{\partial(\rho \bar{q}_z)}{\partial z} = -\frac{\partial \bar{h}_T}{\partial t} \quad (12)$$

Substituting Equations (10) and (11) into (12), the general average Reynolds equation can be obtained and its dimensionless form is:

$$\begin{aligned} & \frac{\partial}{\partial X} \left[\phi_x \frac{\bar{p}}{\eta} H_T^3 \left(\frac{J_1^2}{J_0 - J_2} \right) \frac{\partial \bar{p}}{\partial X} \right] \\ & + \frac{1}{K^2} \frac{\partial}{\partial Z} \left[\phi_z \frac{\bar{p}}{\eta} H_T^3 \left(\frac{J_1^2}{J_0 - J_2} \right) \frac{\partial \bar{p}}{\partial Z} \right] \\ & = -\theta (U_2 - U_1) \frac{\partial}{\partial X} \left[\bar{\rho} \frac{H_T^2}{H_T} \left(1 - \frac{J_1}{J_0} \right) \right] \end{aligned}$$

$$\begin{aligned} & - \frac{\theta}{K} (W_2 - W_1) \frac{\partial}{\partial Z} \left[\bar{\rho} \frac{H_T^2}{H_T} \left(1 - \frac{J_1}{J_0} \right) \right] \\ & - U_1 \frac{\partial}{\partial X} (\bar{\rho} \bar{H}_T) - \frac{\theta}{K} W_1 \frac{\partial}{\partial Z} (\bar{\rho} \bar{H}_T) \\ & - \frac{U_2 - U_1}{2} \bar{\sigma} \frac{\partial(\bar{\rho} \phi_{sx})}{\partial X} \\ & - \frac{W_2 - W_1}{2} \bar{\sigma} \frac{\partial(\bar{\rho} \phi_{sz})}{\partial Z} \quad (13) \end{aligned}$$

where:

$$\begin{aligned} \theta &= \frac{b \eta_0 u_s}{(E' R_x^2)} \\ J_0 &= \int_0^1 F(\tau_e) d\bar{y}, \quad J_1 = \int_0^1 F(\tau_e) \bar{y} d\bar{y}, \\ J_2 &= \int_0^1 F(\tau_e) \bar{y}^2 d\bar{y} \quad (14) \end{aligned}$$

For a Newtonian lubricant, $F(\tau_e) = 1$. Hence, J_0, J_1 and J_2 are equal to 1, 1/2, 1/3 respectively. Substitution into Equation (13), yields the same average Reynolds equation as derived by Zhu and Cheng¹¹.

Elastic deformation

In dimensionless form, the surface elastic deformation is expressed as:

$$\delta(X, Z) = \frac{2}{\pi} \iint_{\Gamma} \frac{\bar{P}(X, Z) + \bar{P}_a(X, Z)}{\sqrt{(X-X')^2 + (Z-Z')^2}} dX' dZ' \quad (15)$$

where Γ is the calculation zone. The mean asperity contact pressure is calculated by the method given by Greenwood and Tripp¹² as:

$$\bar{P}_a = K' F_{2.5}(\lambda) \quad (16)$$

where:

$$K' = \frac{8}{15} \sqrt{2\pi(N\beta\sigma)} \sqrt{\sigma/\beta} \quad (17)$$

and:

$$F_{2.5} = \frac{1}{\sqrt{2\pi}} \int_{\lambda}^{\infty} (s-\lambda)^2 \exp\left(-\frac{s^2}{2}\right) ds \quad (18)$$

The film thickness is given as:

$$H_T = H_0 + \frac{b^2 X^2}{2R_x^2} + \frac{a^2 Z^2}{2R_z R_z} + \delta(X, Z) \quad (19)$$

Viscosity

Among many isothermal viscosity-pressure formulae that have been proposed, the most accurate one for moderate pressure ranges is that of Roelands¹³ and hence it is adopted here:

$$\bar{\eta} = \exp \left\{ (\ln \eta_0 + 9.67) \right. \\ \left. [-1 + (1 + 5.1e^{-9} \cdot E' \cdot \bar{P})^{0.67}] \right\} \quad (20)$$

Load-carrying capacity and traction torque

The total partial EHL load is the sum of the two components: hydrodynamic and asperity contact loads and can be given by:

$$W_T = \iint_{\Gamma} [\bar{P}(X, Y) + \bar{P}_a(X, Y)] dX dY \\ = \iint_{\Gamma} \bar{P}(X, Y) dX dY + \iint_{\Gamma} \bar{P}_a(X, Y) dX dY \quad (21) \\ = W_h + W_a$$

Similarly, the frictional force on the rib face of the inner-ring, F_{rib} also consists of two parts:

$$F_{rib} = F_{r,a} + F_{r,EHL} \quad (22)$$

The first part can be calculated as:

$$F_{r,a} = \mu_a W_a \quad (23)$$

The asperity contact friction coefficient μ_a is equal to 0.2, as suggested by Aihara¹⁴ and Zhou and Hoeprich¹⁵ for metal surfaces. The second part is given by:

$$F_{r,EHL} = abE' \iint_{\Gamma} \bar{\tau}_x|_{Y=0} dX dZ \quad (24)$$

where:

$$\bar{\tau}_x|_{Y=0} = \frac{R_x \theta}{b} (U_2 - U_1) \frac{\bar{\eta}}{H_T J_0} - \frac{R_x}{b} H_T \frac{\partial P}{\partial X} \frac{J_1}{J_0} \quad (25)$$

The total traction torque of the rib to the bearing rotational axis is:

$$M = M_{r,a} + M_{r,EHL} \quad (26)$$

As derived by Aihara¹⁴, the friction torque induced by the asperity contact can be expressed as:

$$M_{r,a} = \frac{\epsilon}{D_a} \cdot F_{r,a} \cdot r_i \quad (27)$$

and similarly:

$$M_{r,EHL} = \frac{\epsilon}{D_a} \cdot abE' \left[\iint_{\Gamma} \bar{\tau}_x|_{Y=0} (a \cdot Z + \epsilon + R_i) \cos \Psi dX dZ \right. \\ \left. - \iint_{\Gamma} \bar{\tau}_Z|_{Y=0} \cdot b \cdot X \cos \Psi dX dZ \right] \quad (28)$$

where:

$$\bar{\tau}_Z|_{Y=0} = \frac{R_x \theta}{b} (W_2 - W_1) \frac{\bar{\eta}}{H_T J_0} \\ - \frac{1}{K} \frac{R_x}{b} H_T \frac{\partial P}{\partial Y} \frac{J_1}{J_0} \quad (29)$$

$R_i = r_i / \cos \Psi$ and D_a is the average diameter of the roller.

Numerical method

For rib/roller end contact in tapered roller bearings, the induced pressure is generally smaller than 0.3 GPa. It is in fact not a heavily loaded situation. Hence, the average Reynolds equation can be solved by the successive over-relaxation (SOR) method and with good convergence. Further, the simultaneous system including the elasticity equation can be solved by the under-relaxation iterative method. The numerical procedure is similar to the one used by Zhu and Cheng¹¹.

Results and discussion

In the present work, a true configuration of an actual tapered roller bearing was used. The selected one was the Chinese made 7518E tapered roller bearing. Its rib/roller end contact is of the tapered rib to spherical end type. According to the result of Jiang *et al.*⁶, this type of configuration comes with the smallest film thickness of all the others. The roughness effects are thus expected to be pronounced. The relevant material and geometrical parameters of the bearing and parameters of a typical bearing lubricant are listed in Table 1. The value of $N\beta\sigma$ is taken as 0.04 as recommended by Zhu and Cheng¹¹. For most engineering surfaces, the value of σ/β is in the range of 0.0001 to 0.01¹¹. In the present analysis, the value of σ/β was chosen to be 0.001 as a reference.

The measured roughness of the rib face is much better than that of the roller big end. This may be due to the fact that grinding the tapered rib surface is much more straightforward. Generally speaking, the roughness of the roller end is in the range of 0.1 to 0.45 μm . Table 2 shows two sets of results calculated for two different values of combined surface roughness, each of which were calculated for three cases: (i) two surfaces having the same roughness; (ii) rough rib face to smooth roller end; (iii) smooth rib face to rough roller end. It can be seen that the effects on film thickness and frictional torque are dominated by the combined roughness σ . The domination of roughness on different surfaces induces only small differences in the results. Hence, for the sake of convenience, the roughnesses of the two contact surfaces were set to be the same in this work. This makes the flow factors ϕ_{sx} and ϕ_{sz} equal to 0 and the governing equations (10) and (11) are thus simplified.

Table 1 Main parameters in the calculation

$E_1 = E_2 = 2.12 \times 10^{11} \text{ n/m}^2$, $\mu_{11} = \mu_2 = 0.3$, $\eta = 2.83 \times 10^{-2} \text{ Pa} \cdot \text{s}$ $r = 0.009739 \text{ m}$, $r_o = 0.0605 \text{ m}$, $r_i = 0.057486 \text{ m}$, $\alpha = 2^\circ$ $\beta_1 = 11.64^\circ$, $\epsilon = 0.002735 \text{ m}$, $N \cdot \beta \cdot \sigma = 0.04$, $\sigma/\beta = 0.001$ $W = 80 \text{ N}$, $n = 1000 \text{ rev/min}$
--

Table 2 Effects of different roughness combinations: $R_{x2}/R_{x1} = 0.8$, $n = 1000$ rev/min, $W = 80$ N

	$\sigma = 0.3536 \mu\text{m}$			$\sigma = 0.4243 \mu\text{m}$		
	$\sigma_1 = \sigma$ $\sigma_2 = 0$	$\sigma_1 = \sigma_2$ $= 0.25 \mu\text{m}$	$\sigma_1 = 0$ $\sigma_2 = \sigma$	$\sigma_1 = \sigma$ $\sigma_2 = 0$	$\sigma_1 = \sigma_2$ $= 0.3 \mu\text{m}$	$\sigma_1 = 0$ $\sigma_2 = \sigma$
$h_m (\mu\text{m})$	0.8281	0.82421	0.8124	0.8504	0.8408	0.8332
$h_c (\mu\text{m})$	0.9315	0.92658	0.9177	0.9531	0.94776	0.9403
$M (\text{N mm})$	1.03	1.076	1.096	2.5	2.553	2.617

The validity of the theoretical treatment has been checked by comparing the numerical results calculated with a very small value of σ to the existing experimental data. Numerical calculations were performed under the same conditions for the experiments as carried out by Gadallah and Dalmaz⁴. Figure 5 shows that the

numerical results correlate well with the experimental data.

In the present analysis, the orientation of roughness is assumed to be isotropic to simplify the consideration of spinning motion in the contact. As results of a typical partial EHL analysis, Fig 6 shows the mean EHL pressure, mean roughness contact pressure and nominal film thickness distribution along the centre axis of the contact ellipse. The effects of roughness are thicker nominal film thickness and smaller mean EHL pressure. In fact, this is an opposite trend to the results obtained by Zhu and Cheng¹¹. This is attributable to the difference in the values of the ellipticity parameter K . In Reference 11, the value of K is about 4 while for a typical rib/roller end elliptical contact, K is nearly 0.35. Thus the contact ellipticity parameters have an influence on the effects of the roughness. This was also observed by Zhu and Cheng¹¹. The small hump at the right bottom of Fig 6 shows the asperity contact pressure. It locates in the minimum film thickness area. As expected, the maximum asperity contact pressure occurs at the minimum film thickness point. It is noted that the asperity contacts happen not only in the lubricated contact region but also at the outlet where the lubrication film ruptures.

The roughness effects on the film thickness and traction torque at moderate speed and load are shown in Fig 7. When $h_{sm}/\sigma > 3$, the roughness effects are small.

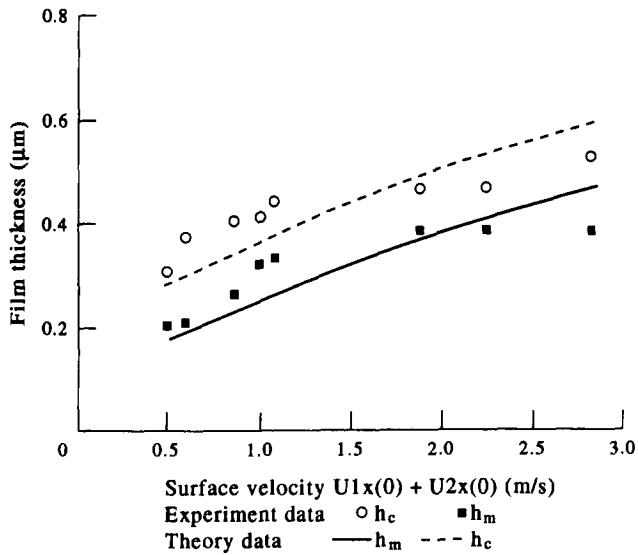


Fig 5 Comparison between theoretical results for smooth contact and Dalmaz experimental data⁴

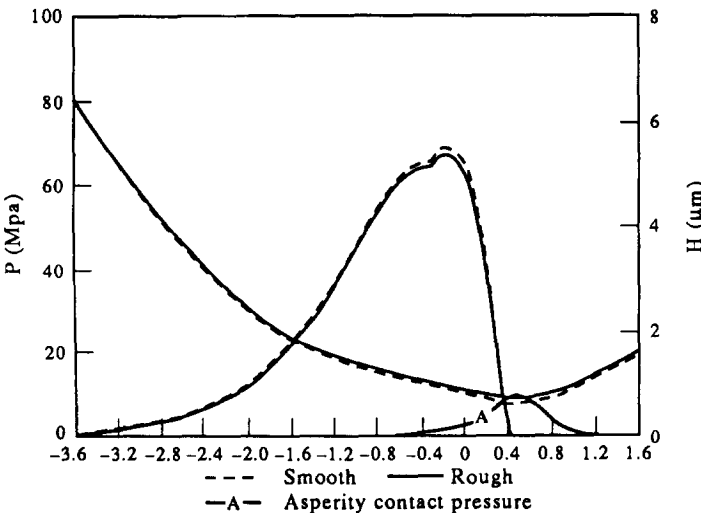


Fig 6 Typical PEHL results for rib-roller end contact at moderate load rotation speed

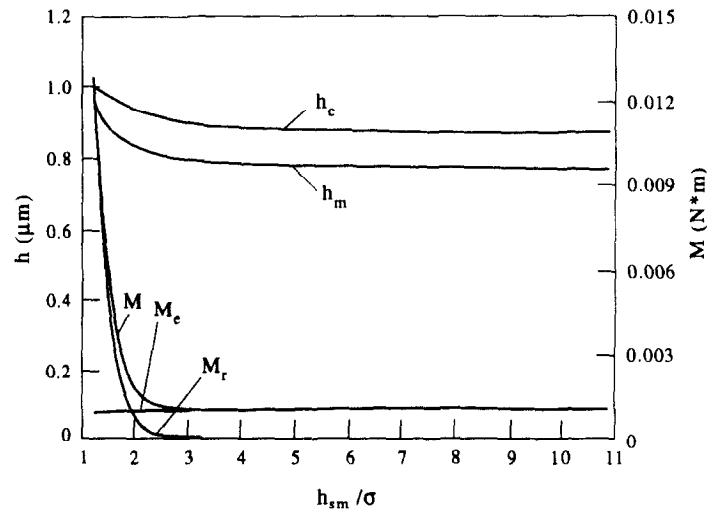


Fig 7 The effects of h_{sm}/σ on friction torque and film thickness

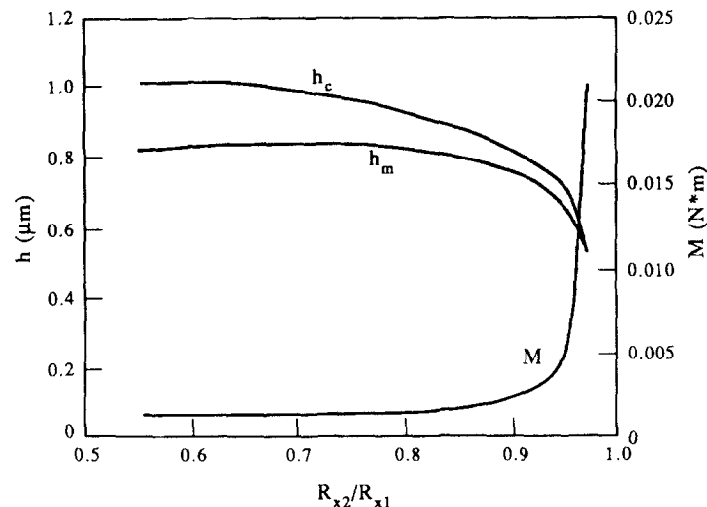


Fig 8 Effects of curvature radius ratio on friction torque and film thickness

While with $h_{sm}/\sigma < 3$, the roughness effects become significant and the total traction torque M , which is the sum of hydrodynamic and asperity contact traction torque, increases rapidly. When $h_{sm}/\sigma < 2$, the total traction is dominated by the asperity contact traction. This leads to the ineffectiveness of lubrication in the contact as a result of the increase in the number of asperity contacts. Hence, a suitable value of h_{sm}/σ must be larger than 2. However, a large value of h_{sm}/σ which means a very smooth surface is not necessary. Figure 7 shows a decreasing trend in film thickness with the increase in h_{sm}/σ values. Therefore, the surface roughness of the roller big end is recommended to be in the range of 0.1 to 0.3 μm for the investigated case, which is equivalent to 5 to 2 in h_{sm}/σ .

Figure 8 shows the effects of the ratios of radius of curvature of roller end to rib face. It is noted that the curves shown in Fig 8 demonstrate the same trends as those in the case of smooth surface contact described by Zhang *et al.*⁵ and Jiang *et al.*⁶. This confirms their finding that the optimal ratios of curvature radius of spherical roller end to tapered rib face is in the range

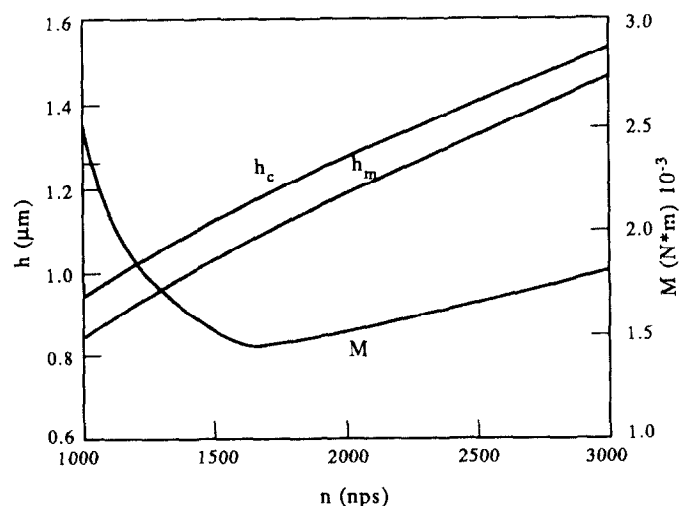


Fig 9 Speed effects on friction torque and film thickness

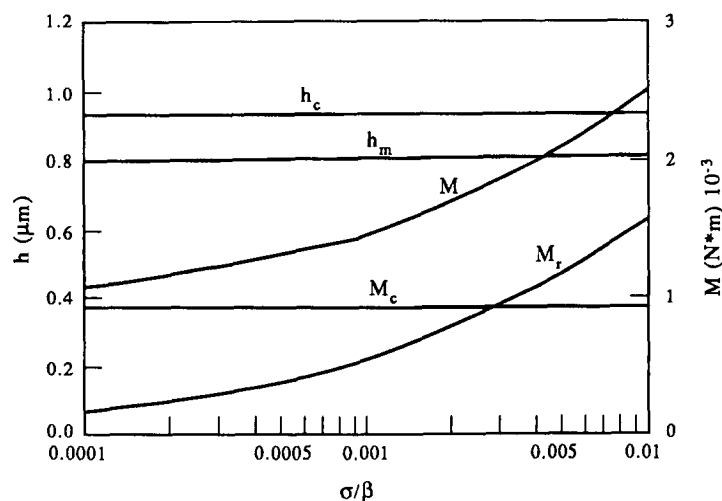


Fig 10 Effects of σ/β on friction torque and film thickness

of 0.6 to 0.8. For values less than 0.6, the radius of the roller big end becomes too small so that it is difficult to control the location of the nominal contact point at the middle of the rib face. Even worse, the contact point may be located in the chamfer or out of rib due to an inaccuracy in manufacturing².

Figure 9 shows the effects of rotation speed on the film thickness and friction torque. Slower speed results in a drop of film thickness and the friction torque is thus increased. For the high speed range, even though the film thickness becomes larger, the total friction torque still goes up slightly due to the increase in the hydrodynamic friction. Hence the optimum speed in the investigated case ought to be 1500 rev/min.

σ/β was taken to be 0.001 in all of the above calculations. In fact, its value is governed by the manufacturing technique employed and is likely to vary from 0.0001 to 0.01¹¹. The effects of σ/β on the film thickness and friction torque are shown in Fig 10. It is seen that when σ/β is larger, the curvature of asperities is smaller for a constant value of surface roughness and thus the asperity contact pressure is higher. This results in a higher friction torque. Its effect on the film thickness is negligible.

Conclusions

A partial EHL analysis of the tapered rib/spherical roller end contact in tapered roller bearings has been completed. By adopting Patir-Cheng's average flow model and Greenwood and Tripp's asperity contact model, the effects of surface roughness on film thickness and friction torque were investigated. Based on the numerical results, the following conclusions can be drawn:

- (1) By choosing a very small value for σ , results of the present analysis for rough surface contact correlate well with the experimental data of smooth surfaces.
- (2) The optimal values of the ratio R_{2x}/R_{1x} for tapered rib/spherical roller contact were deduced to be in the range of 0.6 to 0.8 which confirms findings with the theory of smooth surfaces.

- (3) The effects of surface roughness are significant when $hsm/\sigma < 3$, and become dominant when $hsm/\sigma < 2$. The increase in the amount of asperity contacts leads to ineffectiveness of lubrication such that the friction torque increases drastically. The roughness effects can only be totally neglected when $hsm/\sigma > 5$.
- (4) The optimal surface roughness for the roller big end is in the range of 0.1 μm to 0.3 μm .
- (5) The effect of σ/β on the film thickness is almost negligible but not the friction torque which becomes larger with the increase in σ/β . Hence, appropriate manufacturing techniques should be selected for the finishing of the roller big end surface such that larger curvature of asperities, β , can be achieved.

Acknowledgements

The authors thank the City University of Hong Kong for financial support for the project. Thanks are given to Prof. X. Jiang for his helpful advice.

References

1. Korren H. Gleitreibung und Grenzbelastung an den Bordinflächen von Kegelrollenlagern. *Fortschritt Berichte V.D.I. Zeitschrift*, 1967, Ser. 1, 11
2. Jamison W.E., Kauzlarich J.J. and Mochel E.V. Geometric effect on the rib-roller contact in tapered roller bearings. *ASLE Trans.* 1976, 20, 79-88
3. Dalmaz G., Tessier J.F. and Dudragne G. Friction improvement in cycloidal motion contact: rib-roller end contact in tapered roller bearings. *Proc. 7th Leeds-Lyon Symp. on Tribology. VII (iii)*, 1980, 175
4. Gadallah N. and Dalmaz G. Hydrodynamic lubrication of the rib-roller end contact of a tapered roller bearing. *Trans. ASME J. Tribol* 1984, 106, 265-274
5. Zhang Z., Qiu X. and Hong Y. EHL analysis of rib-roller end contact in tapered roller bearings. *STLE Trans.*, 1988, 31, 461-467
6. Jiang X.F., Wong P.L. and Zhang Z. Thermal non-Newtonian EHL analysis of rib-roller end contact in tapered roller bearings. *ASME/STLE annual Conference, Hawaii, USA October 1994*
7. Kleckner R.J. and Pirvics J. Spherical roller bearing analysis. *ASME Trans. J. Lubric. Technol.* 1982, 104, 99

8. **Patir N. and Cheng H.S.** An average flow model for determining effects of three-dimensional roughness on partial hydrodynamic lubrication. *ASME Trans. J. Lubric. Technol.* 1978, **100**, 12–17
9. **Elrod H.G.** A general theory for laminar lubrication with Reynolds roughness. *ASME Trans. J. Lubr. Technol.* 1979, **101**, 8–14
10. **Tripp J.H.** Surface roughness effects in hydrodynamic lubrication: The flow factor method. *ASME Trans. J. Lubr. Technol.* 1983, **105**, 458–465
11. **Zhu D. and Cheng H.S.** Effect of surface roughness on the point contact EHL. *ASME Trans. J. Tribol.* 1988, **110**, 32–37
12. **Greenwood J.A. and Tripp J.H.** The contact of two nominally flat rough surfaces. *Proc. Inst. Mech. Eng. Part 1*, 1970, **185**, 48, 625–633
13. **Roelands C.J.A.** Correlational aspects of the viscosity–temperature–pressure relationship of lubricating oils. *V.R.B. Groningen, Netherlands*, 1966
14. **Aihara S.** A new running torque formula for tapered roller bearing under axial load. *ASME Trans. J. Tribol.* 1987 **109**, 471–478
15. **Zhou R.S. and Hoeprich M.R.** Torque of tapered roller bearing. *ASME Trans. J. Tribol.* 1991, **113**, 590–597